Chapter Contents

Chapter 12 - Variation Of Parameters

Chapter 11 constructed particular solutions by guessing a derivative-closed trial family. That method is efficient when the equation has constant coefficients and the forcing has one of the familiar polynomial, exponential, or trigonometric forms.

This chapter develops a broader method. Instead of guessing the shape of the forced response, **variation of parameters** begins with the known homogeneous solutions and allows their constant coefficients to become functions.

The main questions are:

- why does replacing constants by functions create a particular solution?
- why is an additional simplifying condition introduced?
- where do the two Wronskian formulas come from?
- why must an equation be put into standard form first?
- how does the method handle forcing terms outside the scope of undetermined coefficients?
- how are initial conditions incorporated without losing the particular solution?
- what changes for first-order and higher-order equations?
- when is variation of parameters theoretically valid but practically awkward?

All examples, equations, and exercises in this chapter are independently constructed. The reference text is used only to identify the mathematical topics.

Turning Homogeneous Constants Into Functions

The Goal Of This Block

The goal is to understand the central move behind variation of parameters:

$$y_h = c_1 y_1 + c_2 y_2$$

becomes the particular-solution proposal:

$$\boxed{y_p = u_1(x)y_1(x) + u_2(x)y_2(x)}.$$

The known homogeneous solutions y_1, y_2 are retained. The constants c_1, c_2 are replaced by unknown functions u_1, u_2 that can adapt to the forcing.

What Problem Are We Solving?

Consider a second-order linear nonhomogeneous equation:

$$a(x)y'' + b(x)y' + c(x)y = g(x),$$

on an interval where $a(x) \neq 0$.

As in Chapter 11, the general solution has the structure:

$$\boxed{y = y_h + y_p},$$

where:

- y_h is the general solution of the associated homogeneous equation
- y_p is one particular solution of the nonhomogeneous equation

Variation of parameters is a method for constructing that one y_p after a fundamental pair of homogeneous solutions is known.

The phrase **fundamental pair** means two linearly independent solutions y_1, y_2. Their linear combinations produce the complete homogeneous family:

$$y_h = c_1 y_1 + c_2 y_2.$$

Why Chapter 11 Needs A Companion Method

Undetermined coefficients depends on predicting a finite trial family that is preserved by differentiation. It works well for forcing terms such as:

$$x^3 - 2x, \qquad e^{4x}, \qquad \cos(5x), \qquad xe^x \sin(2x).$$

Now consider forcing terms such as:

$$\ln x, \qquad \tan x, \qquad \frac{1}{1+x^2}.$$

Repeated differentiation does not keep these functions inside a small, predictable trial family. The Chapter 11 trial ledger therefore has no standard entry for them.

Variation of parameters does not require that kind of guess. It converts the problem into equations for u_1' and u_2', followed by integration.

The practical meaning is:

undetermined coefficients predicts a convenient response shape, whereas variation of parameters builds the response from the homogeneous modes.

Why Vary The Parameters?

Suppose the homogeneous equation has the solution family:

$$y_h = c_1 y_1 + c_2 y_2.$$

The constants c_1, c_2 allow us to select different homogeneous solutions, but constant coefficients cannot create a nonzero forcing. Every such linear combination still satisfies the homogeneous equation.

Variation of parameters replaces the constants by functions:

$$c_1 \longrightarrow u_1(x), \qquad c_2 \longrightarrow u_2(x).$$

This produces:

$$y_p = u_1 y_1 + u_2 y_2.$$

When this expression is differentiated, the new terms u_1' and u_2' appear. Those terms provide the extra freedom needed to reproduce $g(x)$.

Another way to see it:

the homogeneous functions provide the available modes, while the varying coefficients determine how much of each mode is supplied at each value of the independent variable.

Why One Particular Solution Is Still Enough

Suppose variation of parameters produces one function y_p satisfying:

$$L[y_p] = g(x).$$

If integration choices produce another particular solution $\tilde{y}_p$, then:

$$L[\tilde{y}_p] = g(x).$$

Subtract the two equations:

$$\begin{aligned} L[\tilde{y}_p - y_p] &= L[\tilde{y}_p] - L[y_p] \\ &= g(x) - g(x) \\ &= 0. \end{aligned}$$

Thus $\tilde{y}_p - y_p$ is homogeneous. Any difference caused by constants of integration is already represented by y_h.

This is why the integrals used to build u_1, u_2 may be taken without adding arbitrary constants.

Block 1 Summary

Variation of parameters starts from a known fundamental pair y_1, y_2 and seeks:

$$y_p = u_1 y_1 + u_2 y_2.$$

The unknown functions u_1, u_2 vary the homogeneous coefficients so that the new derivative terms can generate the forcing. Only one particular solution is needed because all differences between particular solutions are homogeneous.

Deriving The Second-Order Method

The Goal Of This Block

The goal is to derive, rather than merely quote, the formulas:

$$\boxed{u_1' = -\frac{y_2 R}{W}, \qquad u_2' = \frac{y_1 R}{W}}.$$

Here R is the forcing after the equation has been normalized, and W is the Wronskian of y_1, y_2.

Start With Standard Form

For the derivation, write the equation in **standard form**:

$$y'' + P(x)y' + Q(x)y = R(x).$$

The associated homogeneous equation is:

$$y'' + P(x)y' + Q(x)y = 0.$$

Let y_1, y_2 be linearly independent solutions of this homogeneous equation. Therefore:

$$y_1'' + Py_1' + Qy_1 = 0$$

and:

$$y_2'' + Py_2' + Qy_2 = 0.$$

These two zero identities will be used locally when the proposed particular solution is substituted.

Differentiate The Proposal Without Skipping Terms

Begin with:

$$y_p = u_1 y_1 + u_2 y_2.$$

Differentiate each product separately:

$$\begin{aligned} y_p' &= (u_1 y_1)' + (u_2 y_2)' \\ &= u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2'. \end{aligned}$$

At this point there are two unknown functions but only one differential equation. There is freedom to impose one additional relation that makes the calculation manageable.

Why We Introduce The Auxiliary Condition

Choose the auxiliary condition:

$$\boxed{u_1' y_1 + u_2' y_2 = 0}.$$

This condition is not taken from the original differential equation. It is a deliberate choice that uses the extra freedom in representing one function y_p with two unknown coefficient functions.

Group the full derivative as:

$$y_p' = \underbrace{u_1' y_1 + u_2' y_2}_{0} + u_1 y_1' + u_2 y_2'.$$

The auxiliary condition removes the underbraced group, leaving:

$$\boxed{y_p' = u_1 y_1' + u_2 y_2'}.$$

Why is this useful? If the four-term expression for y_p' were differentiated again, second derivatives u_1'' and u_2'' would appear. The chosen condition prevents those terms from entering the calculation.

What to remember:

the auxiliary condition does not change the differential equation; it selects a convenient pair of coefficient functions representing the particular solution.

Differentiate Again And Substitute

The simplified first derivative is:

$$y_p' = u_1 y_1' + u_2 y_2'.$$

Differentiate both products:

$$\begin{aligned} y_p'' &= (u_1 y_1')' + (u_2 y_2')' \\ &= u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2''. \end{aligned}$$

Now substitute y_p, y_p', and y_p'' into the left side of:

$$y_p'' + P y_p' + Q y_p = R.$$

This gives:

$$\begin{aligned}
&y_p'' + Py_p' + Qy_p \\
&= u_1'y_1' + u_1y_1'' + u_2'y_2' + u_2y_2'' \\
&\quad + P(u_1y_1' + u_2y_2') + Q(u_1y_1 + u_2y_2).
\end{aligned}$$

Group the terms multiplying u_1, the terms multiplying u_2, and the two terms containing derivatives of the parameters:

$$\begin{aligned}
y_p'' + Py_p' + Qy_p = u_1'y_1' &+ u_2'y_2' \\
&+ u_1(y_1'' + Py_1' + Qy_1) \\
&+ u_2(y_2'' + Py_2' + Qy_2).
\end{aligned}$$

Because y_1 and y_2 solve the homogeneous equation:

$$y_1'' + Py_1' + Qy_1 = 0, \qquad y_2'' + Py_2' + Qy_2 = 0.$$

Substitute those two zero values into the grouped expression:

$$y_p'' + Py_p' + Qy_p = u_1'y_1' + u_2'y_2'.$$

The required right side is R, so the remaining equation is:

$$\boxed{u_1'y_1' + u_2'y_2' = R}.$$

The Two-Equation System

The auxiliary condition supplied the first equation:

$$u_1' y_1 + u_2' y_2 = 0.$$

Substitution into the differential equation supplied the second:

$$u_1' y_1' + u_2' y_2' = R.$$

Place them together:

$$\boxed{\begin{aligned} y_1 u_1' + y_2 u_2' &= 0, \\ y_1' u_1' + y_2' u_2' &= R. \end{aligned}}$$

This is an algebraic linear system for the two quantities u_1' and u_2' at each x. The unknowns at this stage are the derivatives of the parameters, not the parameters themselves.

Solve The System With The Wronskian

Write the system in matrix form:

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ R \end{bmatrix}.$$

The determinant of the coefficient matrix is the Wronskian:

$$\boxed{W = y_1 y_2' - y_1' y_2}.$$

Because y_1, y_2 are linearly independent, $W \neq 0$ on the interval under consideration. The system therefore has a unique solution.

For u_1', replace the first column by the right-side column. Cramer's rule gives:

$$\begin{aligned} u_1' &= \frac{\begin{vmatrix} 0 & y_2 \\ R & y_2' \end{vmatrix}}{W} \\ &= \frac{0 \cdot y_2' - y_2 R}{W} \\ &= -\frac{y_2 R}{W}. \end{aligned}$$

For u_2', replace the second column by the right-side column:

$$\begin{aligned} u_2' &= \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & R \end{vmatrix}}{W} \\ &= \frac{y_1 R - 0 \cdot y_1'}{W} \\ &= \frac{y_1 R}{W}. \end{aligned}$$

Thus:

$$\boxed{u_1' = -\frac{y_2 R}{W}, \qquad u_2' = \frac{y_1 R}{W}}.$$

Integrate And Reconstruct The Particular Solution

Once u_1' and u_2' are known, integrate:

$$u_1 = -\int \frac{y_2 R}{W}\, dx, \qquad u_2 = \int \frac{y_1 R}{W}\, dx.$$

Then return to the proposal:

$$y_p = u_1 y_1 + u_2 y_2.$$

Substitute the two integral expressions:

$$\boxed{y_p = -y_1 \int \frac{y_2 R}{W}\, dx + y_2 \int \frac{y_1 R}{W}\, dx}.$$

Do not attach arbitrary constants to these two integrals when seeking one particular solution. If constants k_1, k_2 were included, they would add:

$$k_1 y_1 + k_2 y_2,$$

which is already part of y_h.

Block 2 Summary

For:

$$y'' + Py' + Qy = R,$$

with a fundamental pair y_1, y_2, set:

$$y_p = u_1 y_1 + u_2 y_2.$$

The auxiliary condition and the differential equation produce:

$$\begin{aligned} y_1 u_1' + y_2 u_2' &= 0, \\ y_1' u_1' + y_2' u_2' &= R. \end{aligned}$$

Solving with $W = y_1 y_2' - y_1' y_2$ gives:

$$u_1' = -\frac{y_2 R}{W}, \qquad u_2' = \frac{y_1 R}{W}.$$

Integrate, substitute into $y_p = u_1 y_1 + u_2 y_2$, and finally add y_h.

Standard Form, Intervals, And The Wronskian

The Goal Of This Block

The goal is to prepare an equation correctly before applying the formulas. The most important preliminary step is making the coefficient of y'' equal to 1.

Normalize Every Term

Suppose the equation is:

$$a(x)y'' + b(x)y' + c(x)y = g(x),$$

where $a(x) \neq 0$ on the chosen interval.

Divide **every term** by $a(x)$:

$$\frac{a}{a}y'' + \frac{b}{a}y' + \frac{c}{a}y = \frac{g}{a}.$$

Since $a/a = 1$, the standard form is:

$$y'' + \frac{b}{a}y' + \frac{c}{a}y = \frac{g}{a}.$$

Therefore:

$$P = \frac{b}{a}, \qquad Q = \frac{c}{a}, \qquad \boxed{R = \frac{g}{a}}.$$

The forcing used in the variation formulas is $R = g/a$, not the original g.

The Equivalent Nonstandard Formula

After normalization:

$$R = \frac{g}{a}.$$

Substitute this into the standard formulas:

$$u_1' = -\frac{y_2 R}{W}, \qquad u_2' = \frac{y_1 R}{W}.$$

For u_1':

$$u_1' = -\frac{y_2 (g/a)}{W} = -\frac{y_2 g}{aW}.$$

For u_2':

$$u_2' = \frac{y_1 (g/a)}{W} = \frac{y_1 g}{aW}.$$

Thus the equivalent formulas for the unnormalized equation are:

$$\boxed{u_1' = -\frac{y_2 g}{aW}, \qquad u_2' = \frac{y_1 g}{aW}}.$$

Using these formulas is valid, but normalizing first is usually clearer because it makes the actual forcing R visible.

Compute The Wronskian In A Fixed Order

For an ordered pair (y_1, y_2), use:

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2.$$

If the pair is reversed, then:

$$W(y_2, y_1) = -W(y_1, y_2).$$

The formulas for u_1' and u_2' are tied to the same ordering. Reversing the functions without recomputing the Wronskian is a common source of sign errors.

The Interval Matters

Variation of parameters is applied on an interval where:

- the normalized coefficients P, Q, R are continuous
- the leading coefficient $a(x)$ is nonzero
- the selected fundamental solutions are defined
- the Wronskian is nonzero

For example, if normalization requires division by x^2, the calculation cannot pass through $x = 0$. One might work on $(0, \infty)$ or on $(-\infty, 0)$, but not on a single interval crossing zero.

Likewise, an antiderivative such as:

$$\int \cot x \, dx = \ln|\sin x|$$

must be interpreted on an interval that does not contain a zero of $\sin x$.

A Preparation Example

Prepare the equation:

$$2x^2y'' - 6xy' + 8y = x^4 \ln x, \qquad x > 0.$$

The leading coefficient is:

$$a(x) = 2x^2.$$

Divide each term by $2x^2$:

$$\frac{2x^2}{2x^2}y'' - \frac{6x}{2x^2}y' + \frac{8}{2x^2}y = \frac{x^4 \ln x}{2x^2}.$$

Simplify each coefficient:

$$y'' - \frac{3}{x}y' + \frac{4}{x^2}y = \frac{x^2 \ln x}{2}.$$

Thus the forcing used in the variation formulas is:

$$\boxed{R(x) = \frac{x^2 \ln x}{2}}.$$

The restriction $x > 0$ keeps the divided coefficients defined and makes $\ln x$ real.

Block 3 Summary

Before using variation of parameters:

1. choose an interval on which the equation is regular
2. divide every term by the leading coefficient
3. identify the normalized forcing $R = g/a$
4. order the fundamental solutions and compute $W = y_1y_2' - y_1'y_2$
5. keep that ordering throughout the calculation

The most frequent setup error is using g where the formulas require R.

Complete Second-Order Examples

The Goal Of This Block

The goal is to carry the entire method through two different situations:

- a forcing function outside the usual undetermined-coefficients ledger
- a variable-coefficient equation that must be normalized first

A Trigonometric Forcing Outside The Trial Ledger

Solve on the interval $0 < x < \pi$:

$$y'' + y = \csc x.$$

Step 1: Solve the associated homogeneous equation. The associated homogeneous equation is:

$$y'' + y = 0.$$

Its characteristic equation is:

$$r^2 + 1 = 0.$$

The roots are $r = \pm i$, so choose the ordered fundamental pair:

$$y_1 = \cos x, \qquad y_2 = \sin x.$$

Therefore:

$$y_h = c_1 \cos x + c_2 \sin x.$$

Step 2: Identify the normalized forcing. The coefficient of y'' is already 1, so no division is needed. Therefore:

$$R(x) = \csc x.$$

Step 3: Compute the Wronskian. Differentiate the selected pair:

$$y_1' = -\sin x, \qquad y_2' = \cos x.$$

Use $W = y_1 y_2' - y_1' y_2$:

$$\begin{aligned} W &= (\cos x)(\cos x) - (-\sin x)(\sin x) \\ &= \cos^2 x + \sin^2 x \\ &= 1. \end{aligned}$$

Step 4: Find u_1'. The first variation formula is:

$$u_1' = -\frac{y_2 R}{W}.$$

Substitute $y_2 = \sin x$, $R = \csc x$, and $W = 1$:

$$\begin{aligned} u_1' &= -\frac{(\sin x)(\csc x)}{1} \\ &= -1. \end{aligned}$$

Integrate without adding a homogeneous constant:

$$u_1 = -x.$$

Step 5: Find u_2'. The second variation formula is:

$$u_2' = \frac{y_1 R}{W}.$$

Substitute $y_1 = \cos x$, $R = \csc x$, and $W = 1$:

$$\begin{aligned} u_2' &= \frac{(\cos x)(\csc x)}{1} \\ &= \frac{\cos x}{\sin x} \\ &= \cot x. \end{aligned}$$

On $0 < x < \pi$, $\sin x > 0$, so:

$$u_2 = \int \cot x \, dx = \ln(\sin x).$$

Step 6: Return to $y_p = u_1 y_1 + u_2 y_2$. Restate the reconstruction formula:

$$y_p = u_1 y_1 + u_2 y_2.$$

Substitute $u_1 = -x$, $y_1 = \cos x$, $u_2 = \ln(\sin x)$, and $y_2 = \sin x$:

$$\boxed{y_p = -x\cos x + \sin x \ln(\sin x)}.$$

Step 7: Verify the particular solution directly. Differentiate y_p term by term. First:

$$\frac{d}{dx}(-x\cos x) = -\cos x + x\sin x.$$

For the second product:

$$\begin{aligned}\frac{d}{dx}[\sin x \ln(\sin x)] &= \cos x \ln(\sin x) + \sin x \frac{\cos x}{\sin x} \\ &= \cos x \ln(\sin x) + \cos x.\end{aligned}$$

Add the two derivatives. The $-\cos x$ and $+\cos x$ terms cancel:

$$y_p' = x \sin x + \cos x \ln(\sin x).$$

Differentiate again:

$$y_p'' = \sin x + x \cos x - \sin x \ln(\sin x) + \frac{\cos^2 x}{\sin x}.$$

Now add y_p:

$$\begin{aligned}y_p'' + y_p = \sin x + x \cos x - \sin x \ln(\sin x) + \frac{\cos^2 x}{\sin x} \\ - x \cos x + \sin x \ln(\sin x).\end{aligned}$$

Cancel the opposite terms:

$$y_p'' + y_p = \sin x + \frac{\cos^2 x}{\sin x}.$$

Write $\sin x$ with denominator $\sin x$:

$$\sin x = \frac{\sin^2 x}{\sin x}.$$

Therefore:

$$
\begin{aligned}
y_p'' + y_p &= \frac{\sin^2 x + \cos^2 x}{\sin x} \\
&= \frac{1}{\sin x} \\
&= \csc x.
\end{aligned}
$$

The particular solution is verified.

Step 8: Write the general solution. Combine $y = y_h + y_p$:

$$\boxed{y = c_1 \cos x + c_2 \sin x - x \cos x + \sin x \ln(\sin x), \qquad 0 < x < \pi}.$$

A Variable-Coefficient Equation

Solve for $x > 0$:

$$x^2 y'' - 2xy' + 2y = x^3.$$

Step 1: Normalize the equation. The leading coefficient is x^2. Divide every term by x^2:

$$\frac{x^2}{x^2} y'' - \frac{2x}{x^2} y' + \frac{2}{x^2} y = \frac{x^3}{x^2}.$$

Simplify:

$$y'' - \frac{2}{x} y' + \frac{2}{x^2} y = x.$$

Thus:

$$R(x) = x.$$

Step 2: Solve the Euler-Cauchy homogeneous equation. The associated homogeneous equation before normalization is:

$$x^2y'' - 2xy' + 2y = 0.$$

Try $y = x^m$. Then:

$$y' = mx^{m-1}, \qquad y'' = m(m-1)x^{m-2}.$$

Substitute these expressions:

$$x^2m(m-1)x^{m-2} - 2xmx^{m-1} + 2x^m = 0.$$

Each term contains x^m:

$$[m(m-1) - 2m + 2]x^m = 0.$$

For $x > 0$, divide by x^m:

$$m(m-1) - 2m + 2 = 0.$$

Expand and factor:

$$m^2 - 3m + 2 = 0,$$
$$(m-1)(m-2) = 0.$$

The roots are $m = 1, 2$, so choose:

$$y_1 = x, \qquad y_2 = x^2.$$

Therefore:

$$y_h = c_1 x + c_2 x^2.$$

Step 3: Compute the Wronskian. Differentiate:

$$y_1' = 1, \qquad y_2' = 2x.$$

Then:

$$\begin{aligned} W &= y_1 y_2' - y_1' y_2 \\ &= x(2x) - 1(x^2) \\ &= 2x^2 - x^2 \\ &= x^2. \end{aligned}$$

Since $x > 0$, the Wronskian is nonzero.

Step 4: Find and integrate u_1'. Use:

$$u_1' = -\frac{y_2 R}{W}.$$

Substitute $y_2 = x^2$, $R = x$, and $W = x^2$:

$$\begin{aligned} u_1' &= -\frac{x^2 \cdot x}{x^2} \\ &= -x. \end{aligned}$$

Integrate:

$$u_1 = -\frac{x^2}{2}.$$

Step 5: Find and integrate u_2'. Use:

$$u_2' = \frac{y_1 R}{W}.$$

Substitute $y_1 = x$, $R = x$, and $W = x^2$:

$$\begin{aligned} u_2' &= \frac{x \cdot x}{x^2} \\ &= 1. \end{aligned}$$

Integrate:

$$u_2 = x.$$

Step 6: Reconstruct y_p. Return to:

$$y_p = u_1 y_1 + u_2 y_2.$$

Substitute the four current expressions:

$$\begin{aligned} y_p &= \left(-\frac{x^2}{2}\right)(x) + (x)(x^2) \\ &= -\frac{x^3}{2} + x^3 \\ &= \frac{x^3}{2}. \end{aligned}$$

Step 7: Verify in the original equation. For:

$$y_p = \frac{x^3}{2},$$

the derivatives are:

$$y_p' = \frac{3x^2}{2}, \qquad y_p'' = 3x.$$

Substitute into the original left side:

$$\begin{aligned} x^2 y_p'' - 2x y_p' + 2y_p &= x^2(3x) - 2x\left(\frac{3x^2}{2}\right) + 2\left(\frac{x^3}{2}\right) \\ &= 3x^3 - 3x^3 + x^3 \\ &= x^3. \end{aligned}$$

This matches the original forcing.

Step 8: Write the general solution. Combine y_h and y_p:

$$\boxed{y = c_1 x + c_2 x^2 + \frac{x^3}{2}, \qquad x > 0}.$$

Block 4 Summary

The method does not depend on guessing the particular solution's final form. It depends on four reliable inputs:

1. a normalized equation
2. a fundamental homogeneous pair
3. the Wronskian
4. two integrals for the varying parameters

The reconstruction step must remain explicit:

$$y_p = u_1 y_1 + u_2 y_2.$$

This formula explains exactly where every term in the particular solution comes from.

Definite Integrals And Initial Conditions

The Goal Of This Block

The goal is to use definite integrals to build a particular solution anchored at the same point as the initial conditions. This makes the roles of the forced response and the homogeneous constants especially clear.

Anchoring The Parameters At A Base Point

Choose a base point x_0 inside the working interval. Instead of indefinite integrals, define:

$$u_1(x) = -\int_{x_0}^{x} \frac{y_2(t)R(t)}{W(t)}\,dt$$

and:

$$u_2(x) = \int_{x_0}^{x} \frac{y_1(t)R(t)}{W(t)}\,dt.$$

The letter t is a dummy integration variable. The resulting quantities $u_1(x), u_2(x)$ are functions of the upper limit x.

At $x = x_0$, each integral has equal limits, so:

$$u_1(x_0) = 0, \qquad u_2(x_0) = 0.$$

Because:

$$y_p = u_1 y_1 + u_2 y_2,$$

substitution at the base point gives:

$$\begin{aligned} y_p(x_0) &= u_1(x_0)y_1(x_0) + u_2(x_0)y_2(x_0) \\ &= 0. \end{aligned}$$

Under the auxiliary condition, the first derivative is:

$$y_p' = u_1 y_1' + u_2 y_2'.$$

Therefore:

$$\begin{aligned} y_p'(x_0) &= u_1(x_0)y_1'(x_0) + u_2(x_0)y_2'(x_0) \\ &= 0. \end{aligned}$$

Thus the anchored particular solution satisfies:

$$\boxed{y_p(x_0) = 0, \qquad y_p'(x_0) = 0}.$$

This does not mean every particular solution has zero initial data. It means the chosen lower limits construct one convenient particular solution with that property.

A Forced Initial-Value Problem

Solve on $-\pi/2 < x < \pi/2$:

$$y'' + y = 2\sec x,$$

subject to:

$$y(0) = 3, \qquad y'(0) = -1.$$

Step 1: Find a fundamental homogeneous pair. The associated homogeneous equation is:

$$y'' + y = 0.$$

Choose:

$$y_1 = \cos x, \qquad y_2 = \sin x.$$

Their Wronskian was calculated in Block 4:

$$W = 1.$$

The homogeneous solution is:

$$y_h = c_1 \cos x + c_2 \sin x.$$

Step 2: Build u_1 with lower limit 0. The normalized forcing is:

$$R(x) = 2\sec x.$$

Use the definite-integral formula:

$$u_1(x) = -\int_0^x \frac{y_2(t)R(t)}{W(t)}\,dt.$$

Substitute $y_2(t) = \sin t$, $R(t) = 2\sec t$, and $W(t) = 1$:

$$\begin{aligned} u_1(x) &= -\int_0^x 2\sin t \sec t\,dt \\ &= -2\int_0^x \tan t\,dt. \end{aligned}$$

On the chosen interval, $\cos t > 0$, and:

$$\int \tan t\,dt = -\ln(\cos t).$$

Apply the limits:

$$\begin{aligned} u_1(x) &= -2[-\ln(\cos t)]_0^x \\ &= 2[\ln(\cos t)]_0^x \\ &= 2\ln(\cos x) - 2\ln(1) \\ &= 2\ln(\cos x). \end{aligned}$$

Step 3: Build u_2 with lower limit 0. Use:

$$u_2(x) = \int_0^x \frac{y_1(t)R(t)}{W(t)}\,dt.$$

Substitute $y_1(t) = \cos t$, $R(t) = 2\sec t$, and $W(t) = 1$:

$$\begin{aligned} u_2(x) &= \int_0^x 2\cos t \sec t\,dt \\ &= \int_0^x 2\,dt \\ &= [2t]_0^x \\ &= 2x. \end{aligned}$$

Step 4: Reconstruct the anchored particular solution. Return to:

$$y_p = u_1 y_1 + u_2 y_2.$$

Substitute the four current functions:

$$\boxed{y_p = 2\cos x \ln(\cos x) + 2x \sin x}.$$

At the base point $x = 0$:

$$\begin{aligned} y_p(0) &= 2\cos(0)\ln(\cos 0) + 2(0)\sin(0) \\ &= 2(1)\ln(1) + 0 \\ &= 0. \end{aligned}$$

Step 5: Differentiate the particular solution locally. The auxiliary condition gives the shorter derivative formula:

$$y_p' = u_1 y_1' + u_2 y_2'.$$

Here:

$$y_1' = -\sin x, \qquad y_2' = \cos x.$$

Substitute $u_1 = 2\ln(\cos x)$ and $u_2 = 2x$:

$$y_p' = -2\sin x \ln(\cos x) + 2x \cos x.$$

Evaluate at $x = 0$:

$$\begin{aligned} y_p'(0) &= -2\sin(0)\ln(1) + 2(0)\cos(0) \\ &= 0. \end{aligned}$$

Step 6: Write the complete solution before applying the conditions. Use $y = y_h + y_p$:

$$y = c_1 \cos x + c_2 \sin x + 2\cos x \ln(\cos x) + 2x \sin x.$$

Step 7: Apply $y(0) = 3$. Substitute $x = 0$ into the complete solution:

$$\begin{aligned} 3 &= c_1 \cos(0) + c_2 \sin(0) + y_p(0) \\ &= c_1(1) + c_2(0) + 0 \\ &= c_1. \end{aligned}$$

Therefore:

$$c_1 = 3.$$

Step 8: Apply $y'(0) = -1$. Differentiate the complete solution:

$$y' = -c_1 \sin x + c_2 \cos x + y_p'.$$

Substitute $x = 0$ and $y_p'(0) = 0$:

$$\begin{aligned} -1 &= -c_1 \sin(0) + c_2 \cos(0) + y_p'(0) \\ &= 0 + c_2 + 0. \end{aligned}$$

Thus:

$$c_2 = -1.$$

Step 9: Write the selected solution. Substitute $c_1 = 3$ and $c_2 = -1$ into the complete solution:

$$\boxed{y = 3\cos x - \sin x + 2\cos x \ln(\cos x) + 2x\sin x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}}.$$

Why The Base Point Must Belong To The Interval

The lower limit x_0 must lie inside an interval where the integrands are continuous.

For the equation:

$$y'' + y = \csc x,$$

the forcing is undefined at $x = 0$. Therefore an integral beginning at 0 is not valid for a solution on $(0, \pi)$. A base point such as $x_0 = \pi/2$ would be appropriate on that interval.

The choice of lower limit is mathematical, not cosmetic. It fixes a particular solution and must respect the equation's domain.

Block 5 Summary

Definite integrals anchored at x_0 produce:

$$u_1(x) = -\int_{x_0}^{x} \frac{y_2(t)R(t)}{W(t)}\,dt, \qquad u_2(x) = \int_{x_0}^{x} \frac{y_1(t)R(t)}{W(t)}\,dt.$$

The resulting particular solution satisfies:

$$y_p(x_0) = 0, \qquad y_p'(x_0) = 0.$$

This allows initial data at x_0 to determine the homogeneous constants directly.

First-Order And Higher-Order Versions

The Goal Of This Block

The goal is to see that variation of parameters is not restricted to second-order equations. The same principle works for a single homogeneous mode at first order and for an entire fundamental set at higher order.

First Order: Variation Of A Single Constant

Consider the normalized first-order equation:

$$y' + P(x)y = R(x).$$

Let y_1 be a nonzero solution of the associated homogeneous equation:

$$y_1' + Py_1 = 0.$$

The homogeneous family is:

$$y_h = cy_1.$$

Vary the single constant by proposing:

$$y_p = u(x)y_1(x).$$

Differentiate with the product rule:

$$y_p' = u'y_1 + uy_1'.$$

Substitute y_p and y_p' into the normalized equation:

$$u'y_1 + uy_1' + Puy_1 = R.$$

Group the terms multiplying u:

$$u'y_1 + u(y_1' + Py_1) = R.$$

Since y_1 is homogeneous:

$$y_1' + Py_1 = 0.$$

Substitute this zero value:

$$u'y_1 = R.$$

Divide by the nonzero homogeneous solution y_1:

$$\boxed{u' = \frac{R}{y_1}}.$$

Then:

$$u = \int \frac{R}{y_1}\,dx, \qquad y_p = uy_1.$$

This is the same mechanism that appears in the integrating-factor method, expressed through the homogeneous solution.

A First-Order Worked Example

Solve for $x > 0$:

$$y' + \frac{2}{x}y = x^2.$$

Step 1: Find the homogeneous solution. The associated homogeneous equation is:

$$y' + \frac{2}{x}y = 0.$$

Separate variables:

$$\frac{dy}{dx} = -\frac{2}{x}y.$$

For a nonzero homogeneous solution, divide by y and multiply by dx:

$$\frac{1}{y}\,dy = -\frac{2}{x}\,dx.$$

Integrate:

$$\ln|y| = -2\ln x + C.$$

One convenient nonzero homogeneous solution is:

$$y_1 = x^{-2}.$$

Therefore:

$$y_h = cx^{-2}.$$

Step 2: Vary the constant. Set:

$$y_p = uy_1 = ux^{-2}.$$

The first-order variation formula is:

$$u' = \frac{R}{y_1}.$$

Here $R = x^2$ and $y_1 = x^{-2}$, so:

$$\begin{aligned} u' &= \frac{x^2}{x^{-2}} \\ &= x^{2-(-2)} \\ &= x^4. \end{aligned}$$

Integrate:

$$u = \frac{x^5}{5}.$$

Step 3: Reconstruct the particular solution. Return to $y_p = uy_1$:

$$y_p = \left(\frac{x^5}{5}\right)x^{-2} = \frac{x^3}{5}.$$

Step 4: Write the general solution. Combine the homogeneous and particular parts:

$$\boxed{y = \frac{c}{x^2} + \frac{x^3}{5}, \qquad x > 0}.$$

The Higher-Order System

Consider an nth-order linear equation in standard form:

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = R(x).$$

Suppose $y_1, \dots, y_n$ form a fundamental set for the associated homogeneous equation. Then:

$$y_h = c_1y_1 + \cdots + c_ny_n.$$

Vary all n constants:

$$y_p = u_1y_1 + \cdots + u_ny_n.$$

To avoid higher derivatives of the unknown parameters, impose $n-1$ auxiliary conditions:

$$\begin{aligned} u_1'y_1 + u_2'y_2 + \cdots + u_n'y_n &= 0, \\ u_1'y_1' + u_2'y_2' + \cdots + u_n'y_n' &= 0, \\ &\vdots \\ u_1'y_1^{(n-2)} + \cdots + u_n'y_n^{(n-2)} &= 0. \end{aligned}$$

Substitution into the original equation supplies the final relation:

$$u_1'y_1^{(n-1)} + \cdots + u_n'y_n^{(n-1)} = R.$$

Together these form the Wronskian system:

$$\boxed{\begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \\ \vdots \\ u_n' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ R \end{bmatrix}}.$$

The determinant of the coefficient matrix is the n-function Wronskian. It is nonzero for a fundamental set on the working interval, so the system can be solved for $u_1', \dots, u_n'$.

A Third-Order Worked Example

Solve:

$$y''' - y' = 6x.$$

This problem could be solved more quickly by a simpler method. It is used here because its third-order variation system can be solved transparently.

Step 1: Find a fundamental homogeneous set. The associated homogeneous equation is:

$$y''' - y' = 0.$$

Its characteristic equation is:

$$r^3 - r = 0.$$

Factor:

$$r(r-1)(r+1) = 0.$$

The roots are $0, 1, -1$, so choose:

$$y_1 = 1, \qquad y_2 = e^x, \qquad y_3 = e^{-x}.$$

Therefore:

$$y_h = c_1 + c_2 e^x + c_3 e^{-x}.$$

Step 2: Write the three-equation variation system. For a third-order equation, use two auxiliary equations and one forcing equation:

$$\begin{aligned} y_1 u_1' + y_2 u_2' + y_3 u_3' &= 0, \\ y_1' u_1' + y_2' u_2' + y_3' u_3' &= 0, \\ y_1'' u_1' + y_2'' u_2' + y_3'' u_3' &= R. \end{aligned}$$

The required derivatives are:

	y_1	y_2	y_3
function	1	e^x	e^{-x}
first derivative	0	e^x	$-e^{-x}$
second derivative	0	e^x	e^{-x}

Here $R = 6x$. Substitute the table into the system:

$$\begin{aligned} u_1' + e^x u_2' + e^{-x} u_3' &= 0, \\ e^x u_2' - e^{-x} u_3' &= 0, \\ e^x u_2' + e^{-x} u_3' &= 6x. \end{aligned}$$

Step 3: Solve for u_2' and u_3'. From the second equation:

$$e^x u_2' = e^{-x} u_3'.$$

The two terms on the left of the third equation are therefore equal. Let their common value be q:

$$e^x u_2' = q, \qquad e^{-x} u_3' = q.$$

Substitute these into the third equation:

$$q + q = 6x.$$

Combine the left side:

$$2q = 6x.$$

Divide by 2:

$$q = 3x.$$

Therefore:

$$e^x u_2' = 3x.$$

Multiply both sides by e^{-x}:

$$u_2' = 3xe^{-x}.$$

Similarly:

$$e^{-x} u_3' = 3x.$$

Multiply both sides by e^x:

$$u_3' = 3xe^x.$$

Step 4: Solve for u_1'. Return to the first system equation:

$$u_1' + e^x u_2' + e^{-x} u_3' = 0.$$

The previous step found:

$$e^x u_2' = 3x, \qquad e^{-x} u_3' = 3x.$$

Substitute both values:

$$u_1' + 3x + 3x = 0.$$

Combine the two x-terms:

$$u_1' + 6x = 0.$$

Subtract $6x$ from both sides:

$$u_1' = -6x.$$

Step 5: Integrate the three parameter derivatives. For u_1:

$$u_1 = \int -6x\,dx = -3x^2.$$

For u_2, use integration by parts:

$$u_2 = 3\int xe^{-x}\,dx.$$

Choose:

$$U = x, \qquad dV = e^{-x}\,dx.$$

Then:

$$dU = dx, \qquad V = -e^{-x}.$$

Apply $\int U\,dV = UV - \int V\,dU$:

$$\begin{aligned} u_2 &= 3\left[-xe^{-x} + \int e^{-x}\,dx\right] \\ &= 3[-xe^{-x} - e^{-x}] \\ &= -3(x+1)e^{-x}. \end{aligned}$$

For u_3:

$$u_3 = 3\int xe^x\,dx.$$

Integration by parts gives:

$$\begin{aligned} u_3 &= 3\left[xe^x - \int e^x\,dx\right] \\ &= 3(x-1)e^x. \end{aligned}$$

Step 6: Reconstruct y_p. For third order, the reconstruction formula is:

$$y_p = u_1y_1 + u_2y_2 + u_3y_3.$$

Substitute every current expression:

$$y_p = (-3x^2)(1) + [-3(x+1)e^{-x}]e^x + [3(x-1)e^x]e^{-x}.$$

Use $e^{-x}e^x = 1$ and $e^xe^{-x} = 1$:

$$\begin{aligned} y_p &= -3x^2 - 3(x+1) + 3(x-1) \\ &= -3x^2 - 3x - 3 + 3x - 3 \\ &= -3x^2 - 6. \end{aligned}$$

The constant -6 is a homogeneous solution because constants are multiples of $y_1 = 1$. It can be absorbed into c_1. A simpler representative is therefore:

$$y_p = -3x^2.$$

Step 7: Verify and write the general solution. For $y_p = -3x^2$:

$$y_p' = -6x, \qquad y_p''' = 0.$$

Substitute into the differential operator:

$$\begin{aligned} y_p''' - y_p' &= 0 - (-6x) \\ &= 6x. \end{aligned}$$

The forcing is reproduced. Hence:

$$\boxed{y = c_1 + c_2 e^x + c_3 e^{-x} - 3x^2}.$$

Block 6 Summary

At first order, one constant is varied:

$$y_p = u y_1, \qquad u' = \frac{R}{y_1}.$$

At order n, all n homogeneous coefficients are varied. The first $n-1$ rows of the Wronskian system have right side zero, and the final row has right side R.

The pattern is the same at every order:

start with a fundamental homogeneous set, solve for the derivatives of the varying coefficients, integrate, and reconstruct the particular solution.

Workflow, Method Choice, And Common Mistakes

The Complete Second-Order Workflow

Step 1: Choose a valid interval. Identify zeros of the leading coefficient, singular forcing terms, and any domain restrictions.

Step 2: Normalize the equation. Rewrite the problem as:

$$y'' + Py' + Qy = R.$$

Step 3: Find a fundamental homogeneous pair. Solve the associated homogeneous equation and select an ordered pair y_1, y_2.

Step 4: Compute the Wronskian. Use:

$$W = y_1 y_2' - y_1' y_2.$$

Confirm that $W \neq 0$ on the working interval.

Step 5: Find u_1' and u_2'. Use the normalized forcing R:

$$u_1' = -\frac{y_2 R}{W}, \qquad u_2' = \frac{y_1 R}{W}.$$

Step 6: Integrate separately. Calculate u_1 and u_2. State substitutions, integration by parts, or trigonometric identities used in either integral.

Step 7: Reconstruct y_p. Restate and use:

$$y_p = u_1 y_1 + u_2 y_2.$$

Step 8: Simplify only after reconstruction. Opposite homogeneous terms may cancel, or a homogeneous term may be absorbed into y_h.

Step 9: Form the general solution. Write:

$$y = y_h + y_p.$$

Step 10: Verify and apply extra conditions. Check the particular solution in the original equation. Apply initial or boundary conditions to the complete solution, not to y_p alone.

Choosing Between The Two Particular-Solution Methods

Use **undetermined coefficients** when:

- the equation has constant coefficients
- the forcing belongs to a standard finite derivative-closed family
- a short trial can be written immediately

Use **variation of parameters** when:

- a fundamental homogeneous set is known
- the equation has variable coefficients
- the forcing has no standard undetermined-coefficients trial
- a systematic integral representation is more valuable than a guessed form

If both methods apply, undetermined coefficients is often shorter. Variation of parameters remains valid, but it may create integrals that later simplify back to the same elementary trial.

When The Integrals Do Not Simplify

Variation of parameters can be mathematically successful even when an antiderivative cannot be written with elementary functions.

For example, the method might produce:

$$u_2(x) = \int \frac{e^{x^2}}{1+x^2}\,dx.$$

If no elementary antiderivative is available, the integral expression itself still defines a particular solution on a suitable interval.

Depending on the problem, one may:

- leave the answer in integral form
- express it using a named special function
- approximate the integral numerically

The method has not failed merely because the final integral is not elementary.

Common Mistake: Using The Unnormalized Forcing

For:

$$a(x)y'' + b(x)y' + c(x)y = g(x),$$

the standard-form forcing is:

$$R = \frac{g}{a}.$$

Using g directly in:

$$u_1' = -\frac{y_2 R}{W}, \qquad u_2' = \frac{y_1 R}{W}$$

introduces a missing factor of $1/a(x)$. Divide the entire equation first and write R explicitly before calculating the Wronskian formulas.

Common Mistake: Treating The Auxiliary Condition As A Given Law

The relation:

$$u_1' y_1 + u_2' y_2 = 0$$

is not an additional physical condition and is not copied from the original equation. It is a convenient representation choice that removes u_1'' and u_2'' from the calculation.

Its role should be explained whenever the method is derived, because otherwise the second system equation appears to come from nowhere.

Common Mistake: Losing The Wronskian Sign

With the ordered pair (y_1, y_2):

$$W = y_1 y_2' - y_1' y_2.$$

The matching formulas are:

$$u_1' = -\frac{y_2 R}{W}, \qquad u_2' = \frac{y_1 R}{W}.$$

Write all three formulas together before substitution. Memorizing one minus sign without recording the order of y_1, y_2 is unreliable.

Common Mistake: Adding Integration Constants Twice

If:

$$u_1 = U_1 + k_1, \qquad u_2 = U_2 + k_2,$$

then reconstruction gives:

$$\begin{aligned} y_p &= (U_1 + k_1)y_1 + (U_2 + k_2)y_2 \\ &= U_1y_1 + U_2y_2 + k_1y_1 + k_2y_2. \end{aligned}$$

The last two terms are homogeneous. They duplicate the arbitrary constants in y_h. Omit integration constants while constructing one y_p.

Common Mistake: Forgetting To Return To y_p

Finding u_1 and u_2 does not complete the method. They are coefficient functions, not the particular solution.

Always restate:

$$y_p = u_1y_1 + u_2y_2$$

and substitute all four functions. This prevents u_1 or u_2 from being reported incorrectly as the answer.

Common Mistake: Applying Conditions To The Wrong Function

Initial conditions belong to the complete solution:

$$y = y_h + y_p.$$

Unless a definite-integral construction has deliberately made $y_p(x_0) = y_p'(x_0) = 0$, both the homogeneous and particular parts contribute to the initial values.

Write the full solution and its required derivatives before substituting any condition.

Block 7 Summary

The method is reliable when its decisions occur in the correct order:

1. choose the interval
2. normalize
3. solve the homogeneous problem
4. compute the Wronskian
5. solve for the parameter derivatives
6. integrate
7. reconstruct y_p
8. attach y_h, verify, and apply conditions

Variation of parameters is broader than undetermined coefficients, but broader does not always mean shorter.

Original Practice Set

Practice Problems

Problem 1: Normalize Before Using The Formulas. For $x > 0$, rewrite the equation in standard form and identify P, Q, R:

$$3x^2y'' + 6xy' - 9y = \frac{x^5}{1+x}.$$

Problem 2: Compute A Wronskian. For $x > 0$, calculate $W(y_1, y_2)$ for:

$$y_1 = x^{-1}, \qquad y_2 = x^2.$$

State what the result says about linear independence.

Problem 3: Find The Parameter Derivatives. A standard-form equation has:

$$y_1 = e^x, \qquad y_2 = e^{-x}, \qquad R(x) = \frac{1}{1+x^2}.$$

Calculate W, u_1', and u_2'. Do not evaluate the resulting integrals.

Problem 4: Explain The Anchored Initial Values. Suppose:

$$u_1(x) = -\int_{x_0}^{x} \frac{y_2(t)R(t)}{W(t)}\,dt, \qquad u_2(x) = \int_{x_0}^{x} \frac{y_1(t)R(t)}{W(t)}\,dt.$$

Show that the reconstructed particular solution satisfies:

$$y_p(x_0) = 0, \qquad y_p'(x_0) = 0.$$

Problem 5: Vary One Parameter At First Order. Solve:

$$y' - 3y = e^{2x}$$

by varying the constant in a homogeneous solution.

Problem 6: Build A Third-Order System. A third-order standard-form equation has the fundamental set:

$$y_1 = 1, \qquad y_2 = x, \qquad y_3 = e^x,$$

and normalized forcing:

$$R(x) = \ln x, \qquad x > 0.$$

Write the three equations for u_1', u_2', u_3'. Do not solve them.

Problem 7: Solve A Constant-Coefficient Equation. Use variation of parameters to solve:

$$y'' - 4y = 8e^x.$$

Problem 8: Solve A Nonstandard Trigonometric Forcing. Use variation of parameters to solve on $-\pi/2 < x < \pi/2$:

$$y'' + y = \tan x.$$

Problem 9: Solve A Repeated-Root Euler Equation. Use variation of parameters to solve for $x > 0$:

$$x^2y'' - xy' + y = x^2.$$

Problem 10: Solve An Initial-Value Problem. Use definite integrals based at $x_0 = 0$ to solve:

$$y'' + 4y = 4\sec(2x),$$

subject to:

$$y(0) = 2, \qquad y'(0) = -1,$$

on $-\pi/4 < x < \pi/4$.

Problem 11: Solve A Third-Order Equation. Use variation of parameters to solve:

$$y''' - 4y' = 8.$$

Problem 12: Choose And Diagnose The Method. For each equation, state whether variation of parameters applies and whether undetermined coefficients would be the more efficient method.

(a)

$$y'' + y = xe^x$$

(b)

$$x^2y'' + xy' + y = \ln x, \qquad x > 0$$

(c)

$$y'' + y = e^{x^2}$$

Worked Answers: Problems 1 To 6

Answer 1. Normalize:

$$3x^2y'' + 6xy' - 9y = \frac{x^5}{1+x}, \qquad x > 0.$$

The leading coefficient is $3x^2$. Divide every term by $3x^2$:

$$\frac{3x^2}{3x^2}y'' + \frac{6x}{3x^2}y' - \frac{9}{3x^2}y = \frac{x^5}{3x^2(1+x)}.$$

Simplify the four coefficients:

$$y'' + \frac{2}{x}y' - \frac{3}{x^2}y = \frac{x^3}{3(1+x)}.$$

Comparing with $y'' + Py' + Qy = R$ gives:

$$\boxed{P(x) = \frac{2}{x}, \qquad Q(x) = -\frac{3}{x^2}, \qquad R(x) = \frac{x^3}{3(1+x)}}.$$

Answer 2. The functions are:

$$y_1 = x^{-1}, \qquad y_2 = x^2.$$

Differentiate:

$$y_1' = -x^{-2}, \qquad y_2' = 2x.$$

Use:

$$W = y_1 y_2' - y_1' y_2.$$

Substitute the four expressions:

$$\begin{aligned} W &= (x^{-1})(2x) - (-x^{-2})(x^2) \\ &= 2 - (-1) \\ &= 3. \end{aligned}$$

Since $W = 3 \neq 0$, the two functions are linearly independent on $x > 0$.

Answer 3. The ordered pair and forcing are:

$$y_1 = e^x, \qquad y_2 = e^{-x}, \qquad R = \frac{1}{1+x^2}.$$

Differentiate the pair:

$$y_1' = e^x, \qquad y_2' = -e^{-x}.$$

Compute the Wronskian:

$$\begin{aligned} W &= y_1 y_2' - y_1' y_2 \\ &= e^x(-e^{-x}) - e^x(e^{-x}) \\ &= -1 - 1 \\ &= -2. \end{aligned}$$

For the first parameter derivative, use:

$$u_1' = -\frac{y_2 R}{W}.$$

Substitute $y_2 = e^{-x}$, $R = 1/(1+x^2)$, and $W = -2$:

$$\begin{aligned} u_1' &= -\frac{e^{-x}/(1+x^2)}{-2} \\ &= \frac{e^{-x}}{2(1+x^2)}. \end{aligned}$$

For the second parameter derivative, use:

$$u_2' = \frac{y_1 R}{W}.$$

Substitute $y_1 = e^x$, $R = 1/(1+x^2)$, and $W = -2$:

$$\begin{aligned} u_2' &= \frac{e^x/(1+x^2)}{-2} \\ &= -\frac{e^x}{2(1+x^2)}. \end{aligned}$$

Therefore:

$$\boxed{u_1' = \frac{e^{-x}}{2(1+x^2)}, \qquad u_2' = -\frac{e^{x}}{2(1+x^2)}}.$$

Answer 4. The anchored parameters are:

$$u_1(x) = -\int_{x_0}^{x} \frac{y_2(t)R(t)}{W(t)}\,dt, \qquad u_2(x) = \int_{x_0}^{x} \frac{y_1(t)R(t)}{W(t)}\,dt.$$

At $x = x_0$, both integrals have the same upper and lower limit:

$$u_1(x_0) = 0, \qquad u_2(x_0) = 0.$$

Use the reconstruction formula:

$$y_p = u_1 y_1 + u_2 y_2.$$

Evaluate at x_0:

$$\begin{aligned} y_p(x_0) &= u_1(x_0)y_1(x_0) + u_2(x_0)y_2(x_0) \\ &= 0 + 0 \\ &= 0. \end{aligned}$$

Under the auxiliary condition:

$$y_p' = u_1 y_1' + u_2 y_2'.$$

Evaluate this derivative at x_0:

$$\begin{aligned} y_p'(x_0) &= u_1(x_0)y_1'(x_0) + u_2(x_0)y_2'(x_0) \\ &= 0 + 0 \\ &= 0. \end{aligned}$$

Hence:

$$\boxed{y_p(x_0) = 0, \qquad y_p'(x_0) = 0}.$$

Answer 5. Solve:

$$y' - 3y = e^{2x}.$$

The associated homogeneous equation is:

$$y' - 3y = 0.$$

One nonzero homogeneous solution is:

$$y_1 = e^{3x}.$$

Therefore:

$$y_h = ce^{3x}.$$

Set:

$$y_p = uy_1 = ue^{3x}.$$

For a normalized first-order equation, the variation formula is:

$$u' = \frac{R}{y_1}.$$

Here $R = e^{2x}$ and $y_1 = e^{3x}$, so:

$$\begin{aligned} u' &= \frac{e^{2x}}{e^{3x}} \\ &= e^{-x}. \end{aligned}$$

Integrate:

$$u = -e^{-x}.$$

Return to $y_p = uy_1$:

$$\begin{aligned} y_p &= (-e^{-x})e^{3x} \\ &= -e^{2x}. \end{aligned}$$

Thus:

$$\boxed{y = ce^{3x} - e^{2x}}.$$

Answer 6. The fundamental set and forcing are:

$$y_1 = 1, \qquad y_2 = x, \qquad y_3 = e^x, \qquad R = \ln x.$$

For third order, use:

$$\begin{aligned} y_1u_1' + y_2u_2' + y_3u_3' &= 0, \\ y_1'u_1' + y_2'u_2' + y_3'u_3' &= 0, \\ y_1''u_1' + y_2''u_2' + y_3''u_3' &= R. \end{aligned}$$

The first derivatives are:

$$y_1' = 0, \qquad y_2' = 1, \qquad y_3' = e^x.$$

The second derivatives are:

$$y_1'' = 0, \qquad y_2'' = 0, \qquad y_3'' = e^x.$$

Substitute these values and $R = \ln x$ into the three equations:

$$\boxed{\begin{aligned} u_1' + xu_2' + e^x u_3' &= 0, \\ u_2' + e^x u_3' &= 0, \\ e^x u_3' &= \ln x. \end{aligned}}$$

Worked Answers: Problems 7 To 9

Answer 7. Solve by variation of parameters:

$$y'' - 4y = 8e^x.$$

The associated homogeneous characteristic equation is:

$$r^2 - 4 = 0.$$

Factor:

$$(r-2)(r+2)=0.$$

Choose the ordered fundamental pair:

$$y_1=e^{2x}, \qquad y_2=e^{-2x}.$$

Thus:

$$y_h=c_1e^{2x}+c_2e^{-2x}.$$

Differentiate:

$$y_1'=2e^{2x}, \qquad y_2'=-2e^{-2x}.$$

Compute the Wronskian:

$$\begin{aligned} W &= y_1y_2'-y_1'y_2 \\ &= e^{2x}(-2e^{-2x})-(2e^{2x})e^{-2x} \\ &= -2-2 \\ &= -4. \end{aligned}$$

The normalized forcing is:

$$R=8e^{x}.$$

Use the first formula:

$$u_1'=-\frac{y_2R}{W}.$$

Substitute:

$$\begin{aligned} u_1' &= -\frac{e^{-2x}(8e^x)}{-4} \\ &= 2e^{-x}. \end{aligned}$$

Integrate:

$$u_1 = -2e^{-x}.$$

Use the second formula:

$$u_2' = \frac{y_1 R}{W}.$$

Substitute:

$$\begin{aligned} u_2' &= \frac{e^{2x}(8e^x)}{-4} \\ &= -2e^{3x}. \end{aligned}$$

Integrate:

$$u_2 = -\frac{2}{3}e^{3x}.$$

Return to:

$$y_p = u_1 y_1 + u_2 y_2.$$

Substitute all four functions:

$$\begin{aligned} y_p &= (-2e^{-x})e^{2x} + \left(-\frac{2}{3}e^{3x}\right)e^{-2x} \\ &= -2e^{x} - \frac{2}{3}e^{x} \\ &= -\frac{8}{3}e^{x}. \end{aligned}$$

Therefore:

$$\boxed{y = c_1 e^{2x} + c_2 e^{-2x} - \frac{8}{3}e^{x}}.$$

Answer 8. Solve on $-\pi/2 < x < \pi/2$:

$$y'' + y = \tan x.$$

The associated homogeneous equation has the fundamental pair:

$$y_1 = \cos x, \qquad y_2 = \sin x.$$

Therefore:

$$y_h = c_1 \cos x + c_2 \sin x.$$

The Wronskian is:

$$W = 1.$$

The normalized forcing is:

$$R = \tan x.$$

Use:

$$u_1' = -\frac{y_2 R}{W}.$$

Substitute:

$$\begin{aligned} u_1' &= -\sin x \tan x \\ &= -\frac{\sin^2 x}{\cos x}. \end{aligned}$$

Use $\sin^2 x = 1 - \cos^2 x$:

$$\begin{aligned} u_1' &= -\frac{1 - \cos^2 x}{\cos x} \\ &= -\sec x + \cos x. \end{aligned}$$

Integrate:

$$u_1 = -\ln|\sec x + \tan x| + \sin x.$$

Now use:

$$u_2' = \frac{y_1 R}{W}.$$

Substitute:

$$\begin{aligned} u_2' &= \cos x \tan x \\ &= \sin x. \end{aligned}$$

Integrate:

$$u_2 = -\cos x.$$

Reconstruct the particular solution:

$$\begin{aligned} y_p &= u_1 y_1 + u_2 y_2 \\ &= [-\ln|\sec x + \tan x| + \sin x] \cos x - \cos x \sin x. \end{aligned}$$

The two $\sin x \cos x$ terms cancel:

$$y_p = -\cos x \ln|\sec x + \tan x|.$$

To verify, let:

$$h(x) = \ln|\sec x + \tan x|.$$

On the chosen interval:

$$h'(x) = \sec x.$$

Since $y_p = -\cos x\, h$:

$$\begin{aligned} y_p' &= \sin x\, h - \cos x\, h' \\ &= \sin x\, h - 1. \end{aligned}$$

Differentiate once more:

$$y_p'' = \cos x\, h + \sin x \sec x.$$

Add $y_p = -\cos x\, h$:

$$\begin{aligned} y_p'' + y_p &= \cos x\, h + \sin x \sec x - \cos x\, h \\ &= \sin x \sec x \\ &= \tan x. \end{aligned}$$

Therefore:

$$\boxed{y = c_1 \cos x + c_2 \sin x - \cos x \ln|\sec x + \tan x|}.$$

Answer 9. Solve for $x > 0$:

$$x^2 y'' - xy' + y = x^2.$$

Normalize by dividing every term by x^2:

$$y'' - \frac{1}{x}y' + \frac{1}{x^2}y = 1.$$

Thus:

$$R = 1.$$

For the homogeneous Euler equation, try $y = x^m$. Substitution gives:

$$m(m-1) - m + 1 = 0.$$

Expand:

$$m^2 - 2m + 1 = 0.$$

Factor:

$$(m-1)^2 = 0.$$

The repeated root is $m = 1$, so choose:

$$y_1 = x, \qquad y_2 = x \ln x.$$

Therefore:

$$y_h = c_1 x + c_2 x \ln x.$$

Differentiate the fundamental pair:

$$y_1' = 1, \qquad y_2' = \ln x + 1.$$

Compute the Wronskian:

$$\begin{aligned} W &= x(\ln x + 1) - 1(x \ln x) \\ &= x \ln x + x - x \ln x \\ &= x. \end{aligned}$$

Use:

$$u_1' = -\frac{y_2 R}{W}.$$

Substitute $y_2 = x \ln x$, $R = 1$, and $W = x$:

$$u_1' = -\ln x.$$

Integrate by parts:

$$u_1 = -x \ln x + x.$$

For the second parameter:

$$u_2' = \frac{y_1 R}{W}.$$

Substitute $y_1 = x$, $R = 1$, and $W = x$:

$$u_2' = 1.$$

Integrate:

$$u_2 = x.$$

Reconstruct:

$$\begin{aligned} y_p &= u_1 y_1 + u_2 y_2 \\ &= (-x \ln x + x)x + x(x \ln x) \\ &= -x^2 \ln x + x^2 + x^2 \ln x \\ &= x^2. \end{aligned}$$

Therefore:

$$\boxed{y = c_1 x + c_2 x \ln x + x^2, \qquad x > 0}.$$

Worked Answers: Problems 10 To 12

Answer 10. Solve on $-\pi/4 < x < \pi/4$:

$$y'' + 4y = 4\sec(2x),$$

with:

$$y(0) = 2, \qquad y'(0) = -1.$$

The associated homogeneous equation is:

$$y'' + 4y = 0.$$

Choose:

$$y_1 = \cos(2x), \qquad y_2 = \sin(2x).$$

Thus:

$$y_h = c_1 \cos(2x) + c_2 \sin(2x).$$

Differentiate:

$$y_1' = -2\sin(2x), \qquad y_2' = 2\cos(2x).$$

Compute the Wronskian:

$$\begin{aligned} W &= y_1 y_2' - y_1' y_2 \\ &= 2\cos^2(2x) + 2\sin^2(2x) \\ &= 2. \end{aligned}$$

Use lower limit 0. For u_1:

$$u_1(x) = -\int_0^x \frac{y_2(t)R(t)}{W(t)}\,dt.$$

Substitute $y_2(t) = \sin(2t)$, $R(t) = 4\sec(2t)$, and $W = 2$:

$$\begin{aligned} u_1(x) &= -\int_0^x \frac{4\sin(2t)\sec(2t)}{2}\,dt \\ &= -2\int_0^x \tan(2t)\,dt. \end{aligned}$$

Since:

$$\int \tan(2t)\,dt = -\frac{1}{2}\ln(\cos(2t)),$$

we obtain:

$$\begin{aligned} u_1(x) &= -2\left[-\frac{1}{2}\ln(\cos(2t))\right]_0^x \\ &= [\ln(\cos(2t))]_0^x \\ &= \ln(\cos(2x)). \end{aligned}$$

For u_2:

$$u_2(x) = \int_0^x \frac{y_1(t)R(t)}{W(t)}\,dt.$$

Substitute:

$$\begin{aligned} u_2(x) &= \int_0^x \frac{4\cos(2t)\sec(2t)}{2}\,dt \\ &= \int_0^x 2\,dt \\ &= 2x. \end{aligned}$$

Reconstruct the anchored particular solution:

$$\boxed{y_p = \cos(2x)\ln(\cos(2x)) + 2x\sin(2x)}.$$

The definite limits guarantee:

$$y_p(0) = 0, \qquad y_p'(0) = 0.$$

Write the complete solution:

$$y = c_1\cos(2x) + c_2\sin(2x) + y_p.$$

Apply $y(0) = 2$:

$$\begin{aligned} 2 &= c_1\cos(0) + c_2\sin(0) + y_p(0) \\ &= c_1. \end{aligned}$$

Therefore:

$$c_1 = 2.$$

Differentiate the complete solution:

$$y' = -2c_1 \sin(2x) + 2c_2 \cos(2x) + y_p'.$$

Apply $y'(0) = -1$:

$$\begin{aligned} -1 &= -2c_1 \sin(0) + 2c_2 \cos(0) + y_p'(0) \\ &= 2c_2. \end{aligned}$$

Divide by 2:

$$c_2 = -\frac{1}{2}.$$

Hence:

$$\boxed{y = 2\cos(2x) - \frac{1}{2}\sin(2x) + \cos(2x)\ln(\cos(2x)) + 2x\sin(2x)}.$$

Answer 11. Solve:

$$y''' - 4y' = 8.$$

The associated characteristic equation is:

$$r^3 - 4r = 0.$$

Factor:

$$r(r-2)(r+2)=0.$$

Choose the fundamental set:

$$y_1=1, \qquad y_2=e^{2x}, \qquad y_3=e^{-2x}.$$

Thus:

$$y_h=c_1+c_2e^{2x}+c_3e^{-2x}.$$

For third order, write:

$$\begin{aligned} y_1u_1'+y_2u_2'+y_3u_3'&=0,\\ y_1'u_1'+y_2'u_2'+y_3'u_3'&=0,\\ y_1''u_1'+y_2''u_2'+y_3''u_3'&=R. \end{aligned}$$

Here $R=8$. The needed derivatives are:

	y_1	y_2	y_3
function	1	e^{2x}	e^{-2x}
first derivative	0	$2e^{2x}$	$-2e^{-2x}$
second derivative	0	$4e^{2x}$	$4e^{-2x}$

Substitution gives:

$$\begin{aligned} u_1'+e^{2x}u_2'+e^{-2x}u_3'&=0,\\ 2e^{2x}u_2'-2e^{-2x}u_3'&=0,\\ 4e^{2x}u_2'+4e^{-2x}u_3'&=8. \end{aligned}$$

Divide the second equation by 2:

$$e^{2x}u_2' = e^{-2x}u_3'.$$

Let their common value be q. The third equation becomes:

$$4q + 4q = 8.$$

Combine the left side:

$$8q = 8.$$

Therefore:

$$q = 1.$$

Hence:

$$e^{2x}u_2' = 1, \qquad e^{-2x}u_3' = 1.$$

Solve each equation:

$$u_2' = e^{-2x}, \qquad u_3' = e^{2x}.$$

The first system equation now gives:

$$u_1' + 1 + 1 = 0.$$

Therefore:

$$u_1' = -2.$$

Integrate all three parameter derivatives:

$$u_1 = -2x, \qquad u_2 = -\frac{1}{2}e^{-2x}, \qquad u_3 = \frac{1}{2}e^{2x}.$$

Reconstruct:

$$\begin{aligned} y_p &= u_1y_1 + u_2y_2 + u_3y_3 \\ &= (-2x)(1) + \left(-\frac{1}{2}e^{-2x}\right)e^{2x} + \left(\frac{1}{2}e^{2x}\right)e^{-2x} \\ &= -2x - \frac{1}{2} + \frac{1}{2} \\ &= -2x. \end{aligned}$$

Verify:

$$y_p' = -2, \qquad y_p''' = 0.$$

Then:

$$y_p''' - 4y_p' = 0 - 4(-2) = 8.$$

Therefore:

$$\boxed{y = c_1 + c_2e^{2x} + c_3e^{-2x} - 2x}.$$

Answer 12.

Part (a). The equation is:

$$y'' + y = xe^x.$$

It is linear with constant coefficients, and xe^x belongs to the standard undetermined-coefficients trial family. Variation of parameters applies after the homogeneous pair is found, but undetermined coefficients is likely to be shorter.

Part (b). The equation is:

$$x^2y'' + xy' + y = \ln x, \qquad x > 0.$$

It is linear but has variable coefficients. Undetermined coefficients in its standard constant-coefficient form does not apply. Variation of parameters does apply on $x > 0$ once a fundamental homogeneous pair is known and the equation is normalized by dividing through by x^2.

Part (c). The equation is:

$$y'' + y = e^{x^2}.$$

It is linear and already in standard form. Variation of parameters applies using $y_1 = \cos x$ and $y_2 = \sin x$. Standard undetermined coefficients does not provide a finite trial family for e^{x^2}.

The variation integrals may not have elementary antiderivatives. An integral representation can still be a valid particular solution.

Chapter 12 Final Summary

For a normalized second-order equation:

$$y'' + Py' + Qy = R,$$

begin with an ordered fundamental pair y_1, y_2 and its Wronskian:

$$W = y_1 y_2' - y_1' y_2.$$

Vary the homogeneous parameters:

$$y_p = u_1 y_1 + u_2 y_2.$$

The auxiliary condition and the differential equation produce:

$$\begin{aligned} y_1 u_1' + y_2 u_2' &= 0, \\ y_1' u_1' + y_2' u_2' &= R. \end{aligned}$$

Solving gives:

$$\boxed{u_1' = -\frac{y_2 R}{W}, \qquad u_2' = \frac{y_1 R}{W}}.$$

Integrate and reconstruct:

$$\boxed{y_p = u_1 y_1 + u_2 y_2}.$$

Finally:

$$\boxed{y = y_h + y_p}.$$

The chapter's central lesson is:

normalize first, preserve the order of the homogeneous basis, make the auxiliary condition explicit, integrate the parameter derivatives, and always return to the reconstruction formula before presenting the solution.